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2. EXPERIMENTS ON CISCO PACKET TRACER (SIMULATION TOOL)

Aim:

To understand and study the packet tracer tool installation and users interface overview.

Introduction:

A simulator, as the name suggests, simulates network devices and its environment. Packet tracer is an exciting network design, simulation and modeling tool.

1. It allows you to model complex systems without the need for dedicated equipment.
2. It helps you to practice your network configuration and troubleshooting skills via computer or an Android or iOS based mobile device.
3. It is a available for both linux and windows desktop environments.
4. Protocols in packet tracer are coded to work and behave in the same way as they would on real hardware.

Procedure:

1. From the network component box, click and drag-and-drop the below components

a. 4 Generic PCs and one Hub b. 4 Generic PCs and one switch.

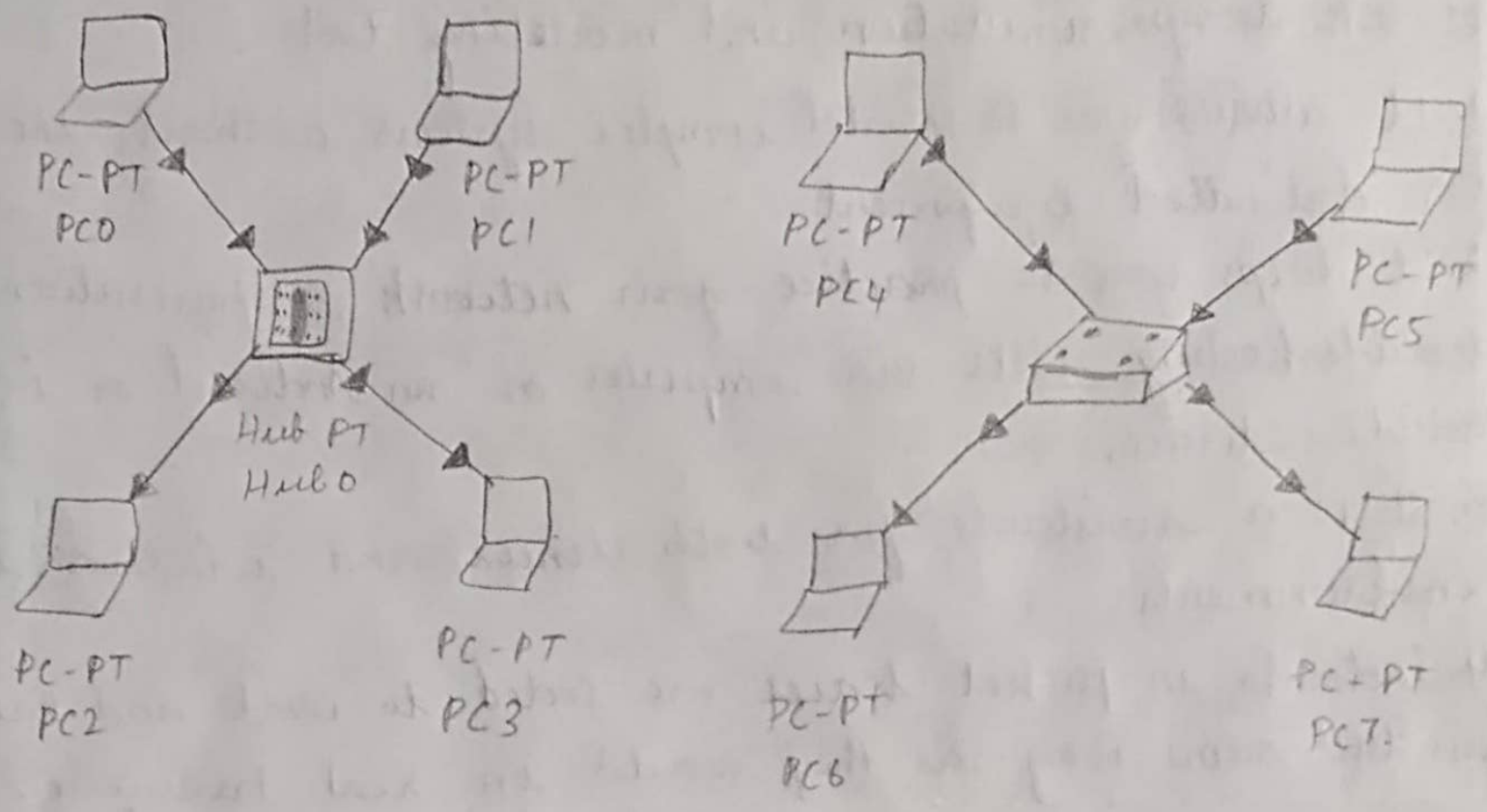
2. Click on connections: a. Click on copper straight-through cable, b. select one of the PC and connect it to hub using the cable. The link LED should glow in green, indicating that the link is up. similarly connect remaining 3 PCs to the hub. c. Similarly connect 4PCs to the switch using copper straight-through cable.

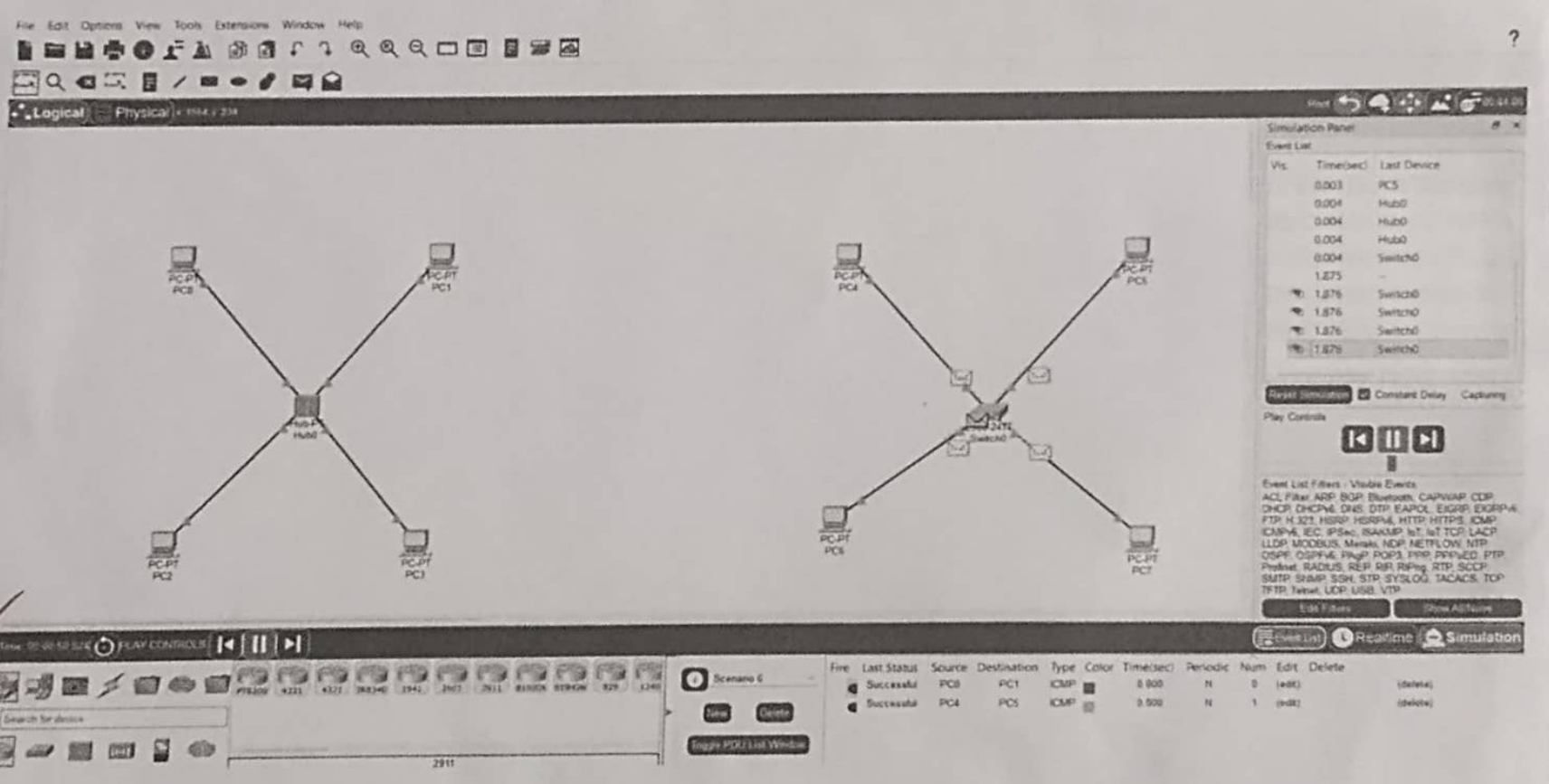
3. Click on the PCs connected to hub, go to the Desktop tab, click on IP configuration, and enter an IP address and subnet mask. Here, the default gateway and DNS Server information is not needed as there are only two end devices in the network.

4. Click on the PDU from the common tool bar,

a. Drag and drop it on one of PC and then drop it on another PC connected to the hub

4. Observe the flow of PDU from source PC to destination PC by selecting the Realtime mode of simulation.
5. Repeat step #3 to step#5 for the PCs connected to the switch.
6. Observe how HUB and switch are forwarding the PDU and write your observation and conclusion about the behaviors of switch and HUB.





Result :

Thus, the network using hub and switch was successfully created and their packet forwarding behaviour was studied using cisco packet tracer.